

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0539

Aes 230 kV
August 25, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0539. The Interconnection Customer proposes a 59.0 MW solar photovoltaic facility interconnecting at the Aes 230 kV substation in the Ohio area. The request was received on November 14, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 217 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$26,043,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

KNOX-AES 230 KV: 113.5% loading (359 MVA) following the outage of the LERDO-TIDD 500 KV L; DFAX = 46.8%. The STATE ST-ROBISON PARK 230 KV is overloaded to 104.6% of its 1,378 MVA emergency rating for the loss of the RICHLAND-DEKALB 500 KV line. The project contributes 6.1% DFAX (3.6 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KNOX-AES 230 KV	Rebuild line with higher-capacity conductor	\$26,008,000	100% (\$26,008,000)
STATE ST-ROBISON PARK 230 KV	Reconductor line	\$3,526,000	1% (\$35,000)
	Total allocated to project		\$26,043,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.